

Climate Shocks and the Credit Market: A Monetary Policy Transmission Lens

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1 Introduction

According to Bernanke and Gertler (1995), the credit channel is one of the key monetary policy transmission channels. Monetary policy affects the credit channel through its impact on the external finance premium ¹. Two linkages underscore monetary policy impacts on this premium: the balance sheet and the bank lending channel. The balance sheet channel stems from dynamics in the balance sheet and the incomes of borrowers, while the lending channel deals with how interest rates could impact lenders' loan supply volumes (Hall, 2001). The balance sheet channel has been well established as a significant monetary policy transmission channel (Aysun and Hepp, 2013; Altavilla et al., 2024; Ashcraft and Campello, 2007). Notwithstanding, there remains room to provide empirical estimations of the credit channel's strength due to recent developments in the identification of monetary policy shocks. This is especially true for emerging economies where this work is scant (Pirozhkova and Viegi, 2025). This paper hones in on the balance sheet channel by estimating how climate shocks can reduce (or indeed enhance) the effectiveness of unanticipated monetary policy decisions on the behaviour of borrowers.

The main motivation of this paper is that like monetary policy shocks, climate shocks – because of their impacts on capital and labour inputs and, in turn, their erosion of firm profitability – can cause frictions within the credit market and have consequential financial stability ramifications (Dafermos et al., 2018; Aglietta and Espagne, 2016). In so doing, climate shocks could widen the external finance premium and, in turn, stymie the efficacy of monetary policy pass-through. While this seems a likely course, there is also the potential for monetary policy to mute the deleterious climate shock effects on the credit market, in line with the literature that studies how monetary policy can be used as a tool to tackle climate change (Roy, 2024; Monasterolo and Raberto, 2018). In either case, climate shocks play a role in the functioning of the credit market and, consequently, the credit market monetary transmission channel.

I will investigate to what extent the manifestation of climate shocks act as a credit market

¹The external finance premium is governed by credit-market frictions which cause departures from lenders expected return and potential borrowers' expected costs.

friction and how this (potential) friction impacts monetary policy transmission. This involves three parts. First, I will compute the monetary policy shock variable. Second, I will estimate the causal relationships of interest in this work using a panel regression. Finally, I will use a quantile regression procedure to nuance these results.

Only physical shocks, not transition

2 Literature review

3 Data and methodology

3.1 Data

The analysis draws on three main data sources: climate shocks, monetary policy shocks, and credit extension data. The sample covers South Africa over the period 2011–2025. All series are observed at a quarterly frequency, determined by the availability of the monetary policy shock data; higher-frequency series are aggregated to quarterly values by taking averages or sums where appropriate.

Climate data: Temperature and precipitation data are sourced from the ERA5 reanalysis dataset produced by the European Centre for Medium-Range Weather Forecasts. ERA5 provides globally gridded, hourly atmospheric estimates at a 31 km horizontal resolution. Grid-level values are weighted by population and aggregated to monthly averages over South Africa’s land area, from which we construct climate shock variables that capture deviations from long-run historical norms in temperature and precipitation.

The climate shocks are calculated as follows:

$$W_t = \frac{T_t - \bar{T}_t}{\sigma(T_t)}, \quad (1)$$

where W_t is the temperature or precipitation shock at time t and T_t is the observed population-weighted climate variable. For both temperature and precipitation, $\bar{T}_t$ is the calendar-month mean computed over the full sample period (e.g., the mean of all January values from 2011 to

2025), and $\sigma(T_t)$ is the corresponding within-month standard deviation. The resulting shock series are shown in Figure A3 in the Appendix.

Monetary policy data: The monetary policy shock series is constructed using high-frequency policy surprises for South Africa from Pirozhkova and Viegi (2025), alongside the repurchase (repo) rate from the South African Reserve Bank (SARB). The high-frequency surprises serve as an instrument to purge the endogenous component of policy rate changes, yielding a shock series that satisfies the exogeneity requirement of the regression framework described in Section 3.2. For robustness, we consider both market-based shocks from Pirozhkova and Viegi (2025) (see Figure A4) and a theory-based shock derived from the policy rate, as described in Section 3.2 below (see Figure A5). Surprises are timed to coincide with policy rate announcement dates and aggregated to a quarterly frequency.

Credit extension data: Private sector credit extension data — disaggregated by borrower type (households and corporates) at the bank level — are obtained from the SARB’s BA900 and BA930 bank compliance reports.² These series include both credit volumes and lending rates, and are constructed following Sibande et al. (2025) and Sibande and Milne (2025) by sorting and aggregating individual bank records into the broad credit categories used in the study: unsecured, secured, and mortgage credit. Summary charts are provided in Figure A1 and Figure A2.

3.2 Methodology

We use a monetary policy shock, rather than the monetary policy rate itself, as an indicator for monetary policy to ensure exogeneity within the regression framework (Miyajima, 2024). To this end, I replicate a shock variable devised by Merrino (2021) for the South African case based on a simplification of the work of Romer and Romer (2004) as follows:

$$\Delta r_t = \alpha + \tilde{r}_t + \sum_{t=0}^2 \beta_t \Delta \tilde{y}_t + \sum_{t=0}^2 \delta_t \tilde{\pi}_t + \varepsilon_t, \quad (2)$$

where Δr_t is the change in the forecast of the repurchase rate, $\tilde{r}_t$ is the forecasted repo rate

²BA930 data at the bank level are not publicly available.

in levels, $\tilde{y}_t$ is the forecasted real GDP growth rate, and $\tilde{\pi}_t$ is the forecasted consumer price index inflation rate. The residuals from Equation 2, ε_t , are used as the MPShock $_t$ explanatory variable that follows in the panel regression estimation. We use a panel regression to primarily test whether monetary policy shocks, when augmented with climate shocks, yield heightened or muted impacts on the credit market as follows:

$$\text{Credit}_{i,t} = \beta_0 + \beta_1 \text{Climate}_t^k + \beta_2 \text{MPShock}_t + \beta_3 (\text{Climate}_t^k \times \text{MPShock}_t) + \Gamma X_t + \gamma_s + \delta_t + \varepsilon_{i,t}, \quad (3)$$

where Credit $_{i,t}$ is the growth rate of credit extended to agent (s) (households or corporates) at time (t), Climate $_t^k$ is the climate shock variable, including temperature and precipitation shocks, as well as extreme wet and dry events, which are each denoted by k . The Climate $_t^k \times \text{MPShock}_t$ term tests whether the frictions introduced by climate change shocks are significant enough to modify the pass-through of monetary policy through the credit channel. X_t is a vector of control variables that includes GDP growth, the US Dollar-Rand exchange rate, the loan-to-deposit ratio and the unemployment rate. γ_s are agent, time-invariant fixed effects, δ_t are time fixed effects, and $\varepsilon_{i,t}$ is the error term.

- robustness specification
- local projections specification

4 Results

5 Drivers

5.1 Timing sensitivity

5.2 Sectoral sensitivity

5.3 Type of shock

6 Conclusion

References

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A Appendix

A.1 Data

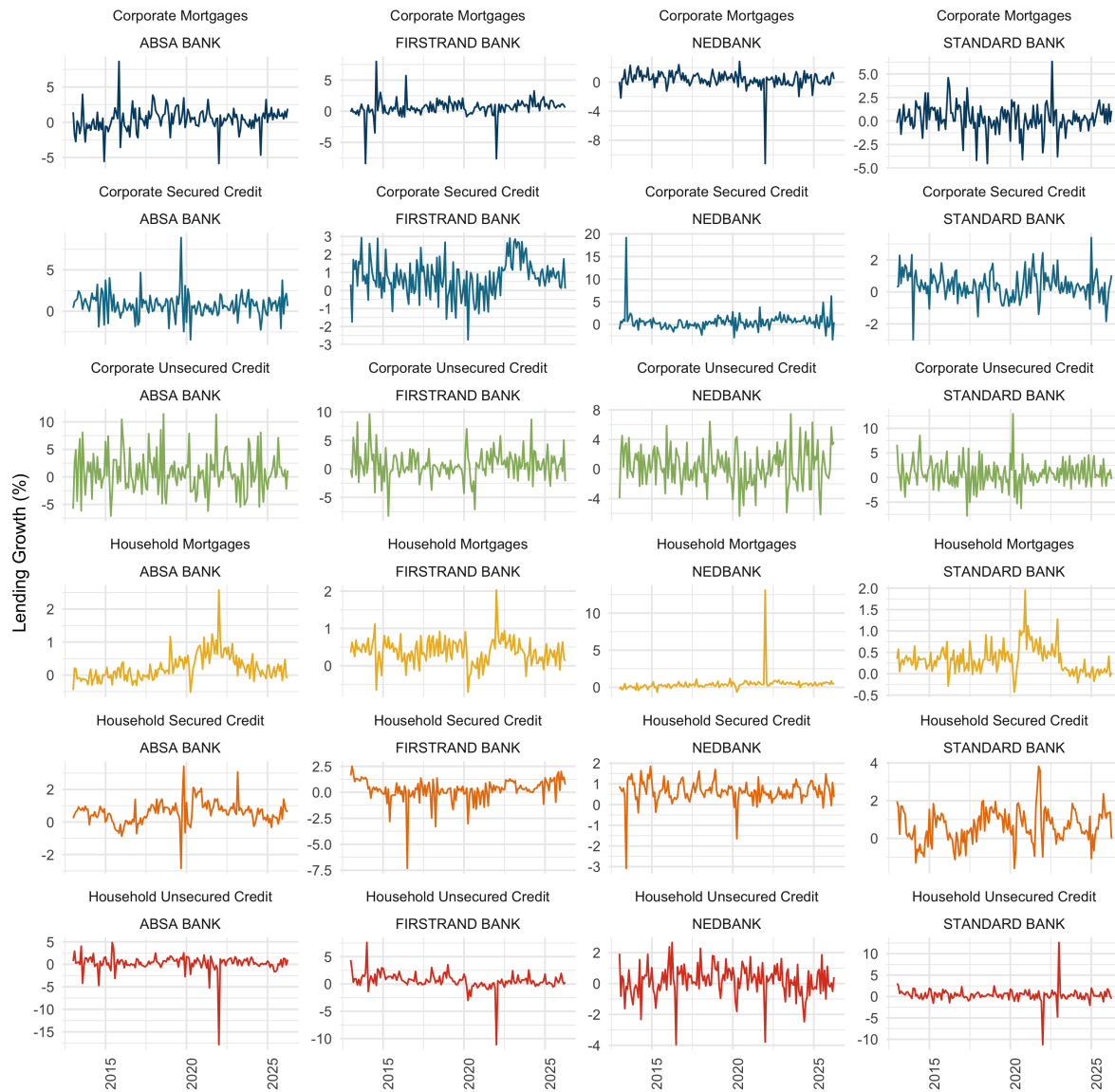


Figure A1: Lending volume growth over time

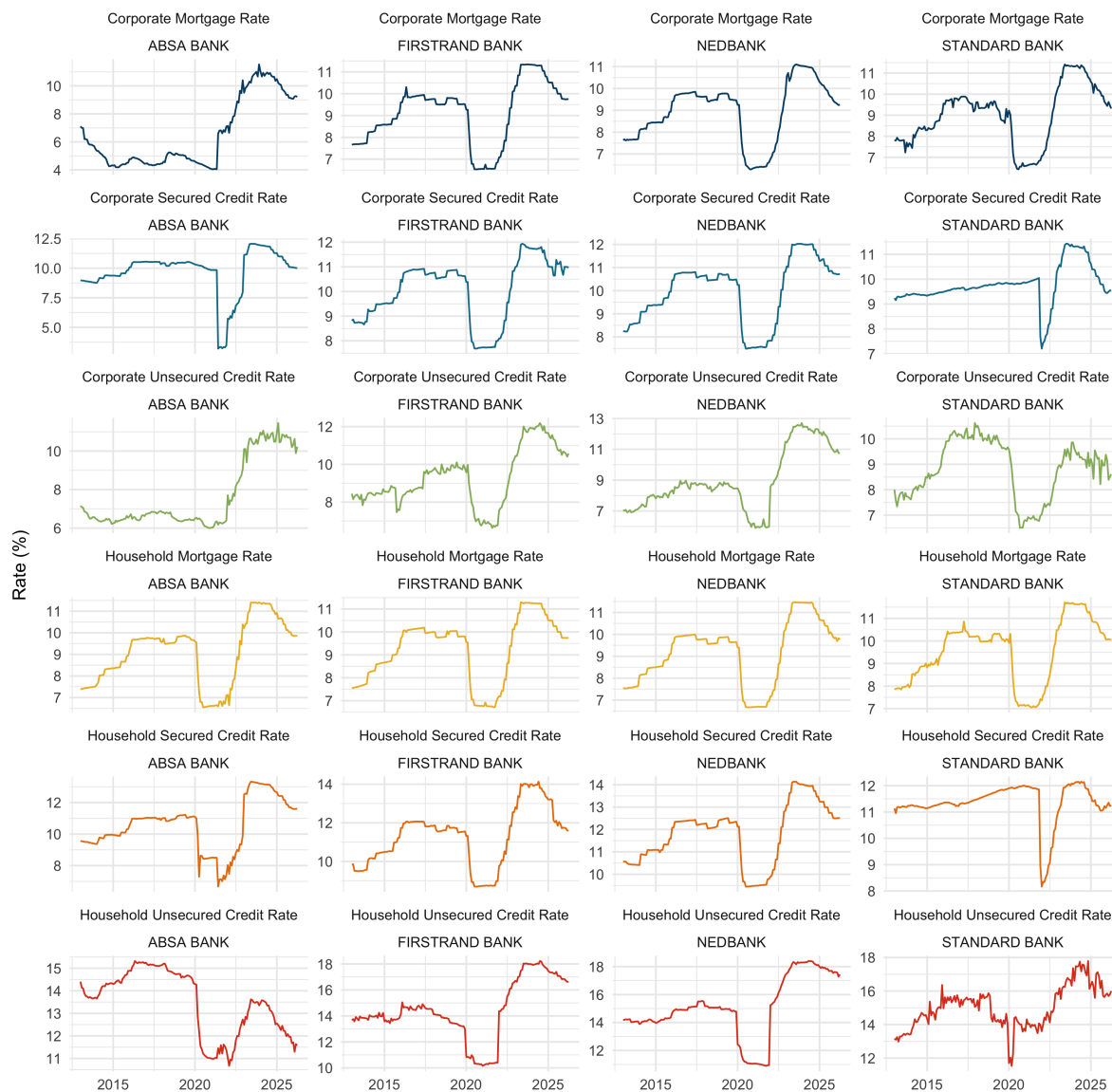


Figure A2: Lending rates overtime

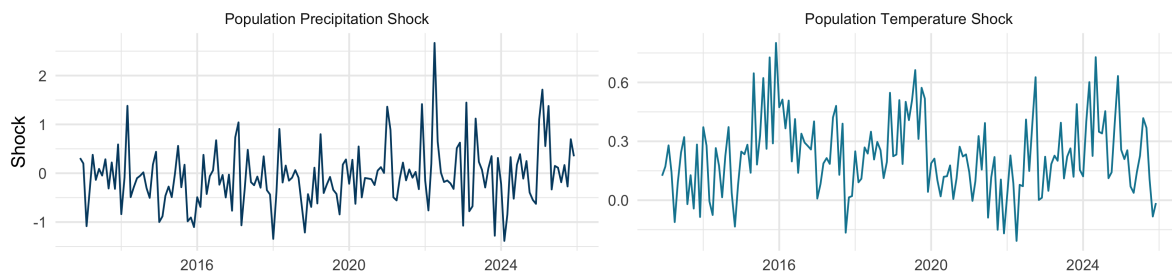


Figure A3: Climate Shocks Data

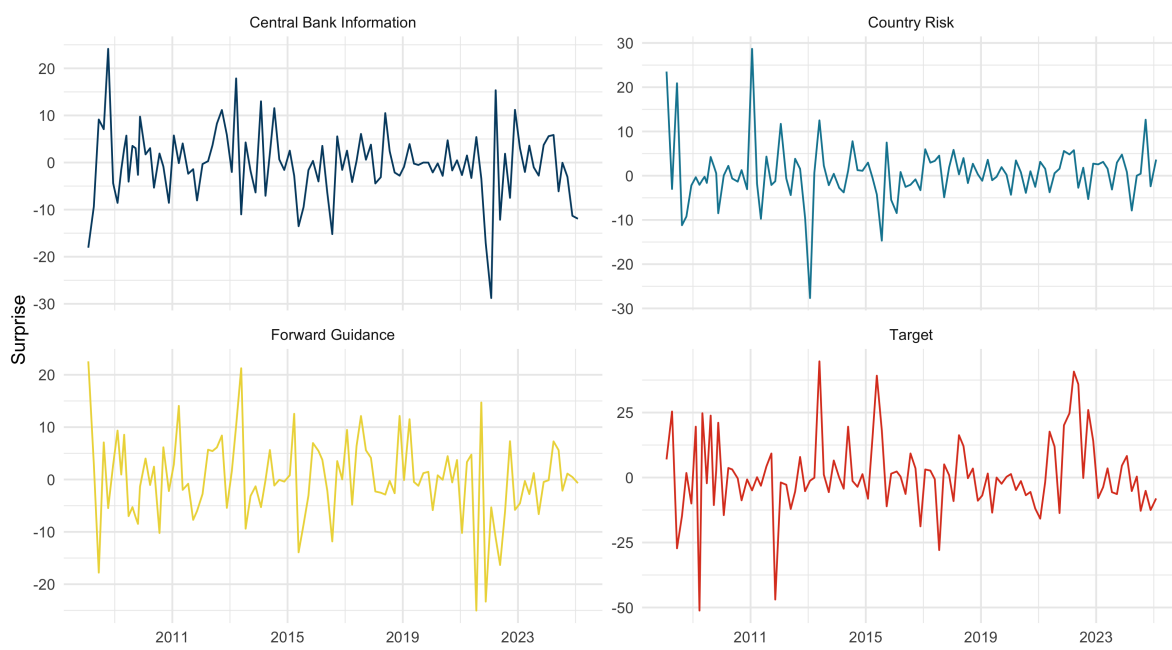


Figure A4: Market-Based Surprises Data

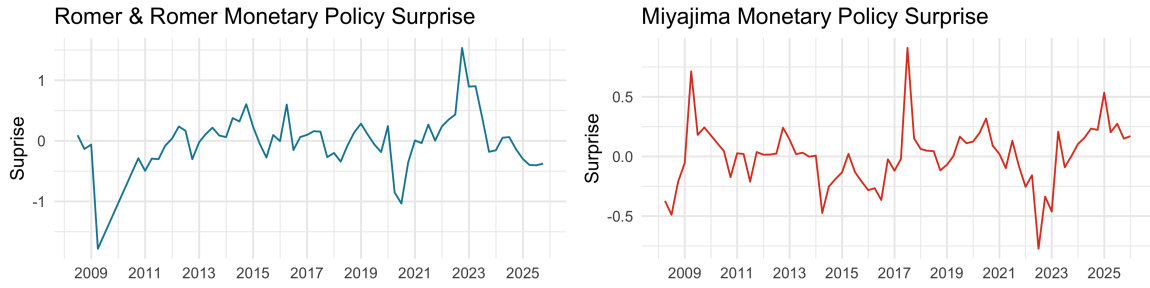


Figure A5: Theory-based surprises data

A.2 Descriptive Statistics

Table A1: Descriptive statistics

	Variable	Mean	Median	SD	Min	Max	N
Monetary Policy	Central Bank Information	-1.64	-0.42	5.69	-17.14	11.19	192
	Country Risk	0.64	0.47	3.38	-5.47	12.50	192
	Forward Guidance	-0.25	-0.47	6.93	-23.32	21.24	192
	Miyajima Surprise	-0.00	0.01	0.26	-0.77	0.91	208
	Romer Surprise	0.06	0.06	0.41	-1.04	1.53	204
	Target	3.12	0.64	13.46	-18.78	44.70	192
Corporate Credit	Corporate Mortgage Rate	8.43	9.07	2.04	4.06	11.35	212
	Corporate Mortgages	74.39	75.79	6.99	24.93	77.82	212
	Corporate Secured Credit	71.57	72.62	6.62	24.16	74.80	212
	Corporate Secured Credit Rate	9.91	9.99	1.36	3.30	12.06	212
	Corporate Unsecured Credit	76.25	77.16	7.01	25.81	79.22	212
	Corporate Unsecured Credit Rate	8.71	8.62	1.74	5.95	12.59	212
Household Credit	Household Mortgage Rate	9.30	9.75	1.41	6.56	11.67	212
	Household Mortgages	77.38	78.22	7.16	26.14	80.28	212
	Household Secured Credit	73.94	75.28	6.87	24.99	77.44	212
	Household Secured Credit Rate	11.28	11.35	1.42	7.09	14.11	212
	Household Unsecured Credit	73.93	75.12	6.86	24.58	76.35	212
	Household Unsecured Credit Rate	14.52	14.54	1.96	10.22	18.37	212
	Population Precipitation Shock	-0.16	-0.16	1.16	-2.99	3.37	204
	Population Temperature Shock	0.23	0.25	0.15	-0.07	0.61	204

A.3 Sector Credit Exposure

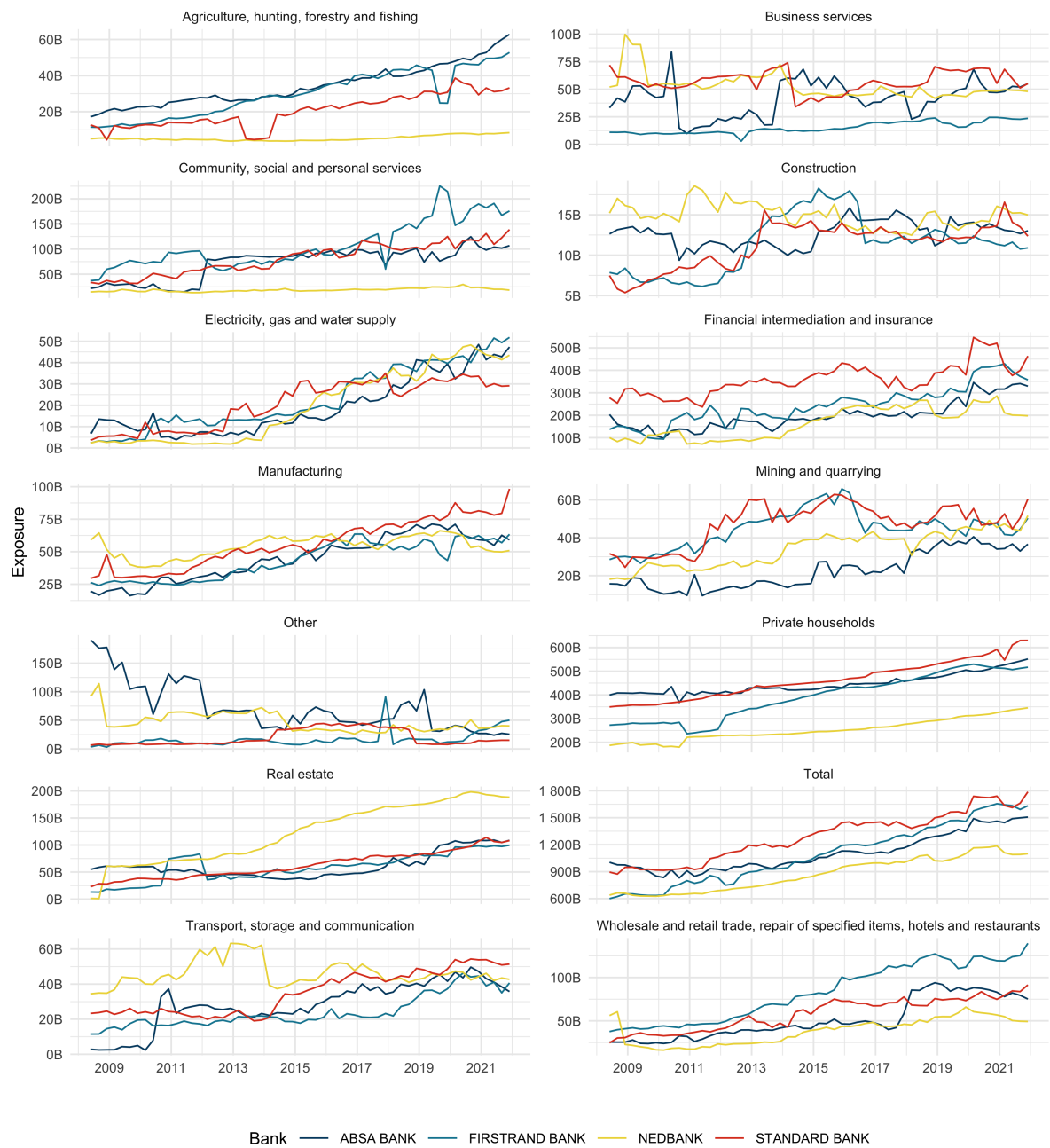


Figure A6: Sector credit exposure